

This deck contains the current thesis material on east–north balance, regional changes, terrain, equipment and wind.

Directional structure and environment

Version: 14/09/2026

Chapter 3, Sections 3.5.1–3.5.3

Thesis §3.5.1

East–north balance in the retained archive

$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1 / \lambda_2 = 19.$$

This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

Equal east and north moments do not establish isotropy

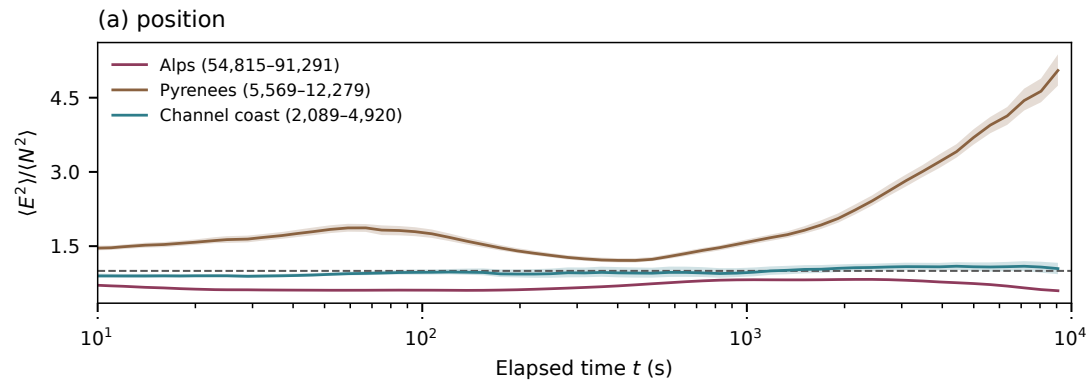
Discussion. Are we observing mean drift, centred directional variability, or a mixture of individually oriented routes?

Full isotropy requires rotation invariance of the law. An isotropic covariance alone would still be only a second-order condition. Raw moments additionally contain the outer product of the mean with itself. An intentional downwind route can be successful despite being directional, so this ratio is not a direct performance score.

Regional position moments differ along east and north

Discussion. How much imbalance reflects mean drift, route orientation or changing support?

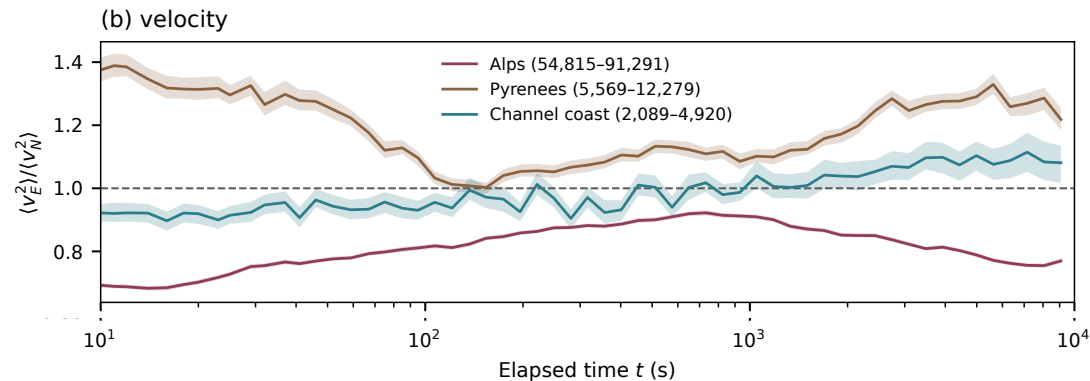
The clock is time since estimated launch, not an increment lag. Position ratios approach five in the Pyrenees. Raw second moments include both covariance and the squared mean. Splits prevent cross-gap stencils but recorded endpoint offsets can still affect these launch-referenced coordinates.



Regional velocity moments show a related directional contrast

Discussion. Which position and velocity contrasts persist under matched conditions?

The regional figure pools paired east/north observations at each elapsed time and uses site-day pointwise bands. Terrain, weather, routes and the disappearing short-flight population can all contribute. Coordinate-moment balance is weaker than full isotropy.



The ratios use paired east and north observations. A site–day bootstrap with 200 draws gives pointwise 10–90% ranges; each plotted time needs at least 30 flights and 10 clusters.

Missing site/date keys are treated as singletons. Resampling these groups does not match the equipment populations on weather, task or duration.

Regional uncertainty bands do not match the groups

Discussion. Would a paired site–day comparison change the group difference and its uncertainty?

The bands are not 95-percent intervals, simultaneous bands or direct intervals for the difference between groups. Their overlap is not itself a calibrated significance test. The many displayed time points are correlated and must not be counted as independent confirmations of an ordering.

Thesis §3.5.2

Regional principal axes

A rotation preserves distance

$$\Sigma = U\Lambda U^\top, \quad |U^\top \mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

The following panels report regional PCA and its support. Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

Ground velocity has two unknown terms

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

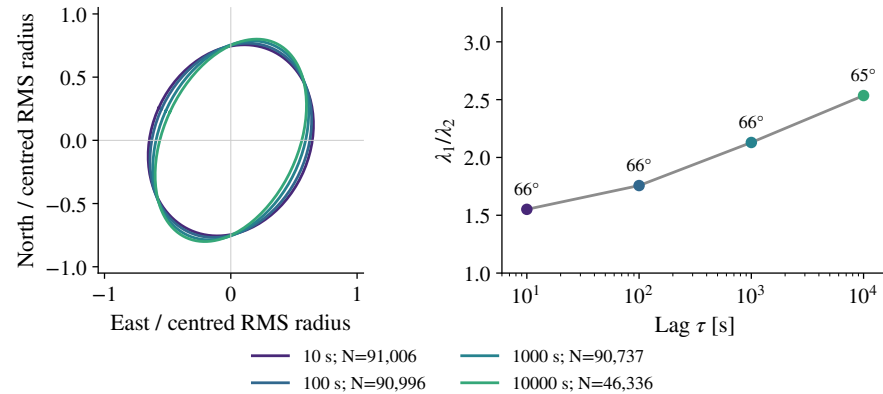
Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

Different marginal exponents, or changing joint dependence, can prevent radial scaling.

PCA describes preferred axes; wind needs independent information

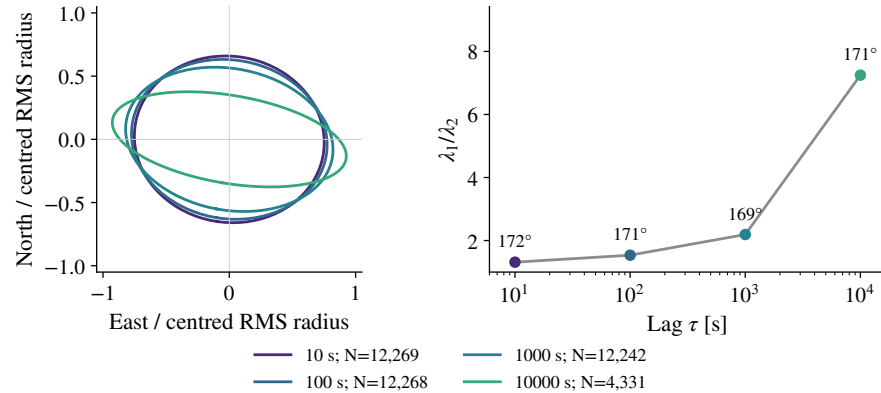
Discussion. Choose matched environments and an independent wind estimate before defining an “environment-free” model target.

Even independent Gaussian components can yield a changing radial exponent if their component exponents differ. If both component exponents agree, changing dependence can still prevent joint scaling. Conversely, a joint vector scaling law always implies radial scaling, whether the vector is isotropic or not. Rotating each flight randomly would balance aggregate directions without changing radial displacements; it cannot recover still-air motion. Independent wind information or carefully justified phase-based wind estimates should be discussed before choosing an environmental correction.

**Alps: centred principal axes across four lags**

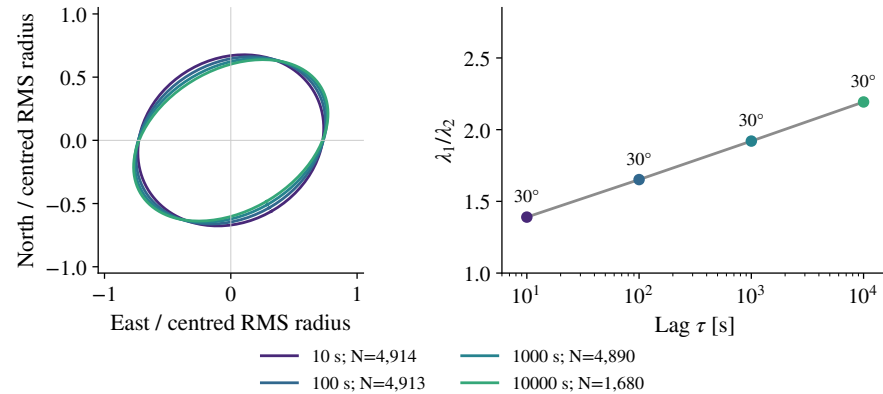
Discussion. Does the axis remain stable as lag and contributing population change?

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

**Pyrenees: centred principal axes across four lags**

Discussion. How much of the stronger 10,000-s elongation persists when flight identities are held fixed?

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

**Channel Coast: centred principal axes across four lags**

Discussion. Does the axis remain stable as lag and contributing population change?

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

NOAA ETOPO 2022

Same Alpine and Pyrenean take-off boxes.

One-arc-minute elevation sampling; retain the footprint above 1000 m.

Project cell centres to regional east/north coordinates. The area-weighted spatial covariance gives the main axis.

NOAA NCEI, ETOPO 2022. Both elevation subsets, requests, code and hashes are archived. Angles are counterclockwise from east, modulo 180°.

Definition sensitivity

Thresholds: 500, 1000, 1500 m.

Alps: 42.1–47.9°.

Pyrenees: 169.7–175.4°.

These spans are not confidence intervals; the axis describes an area, not a traced crest.

A terrain direction measured independently of the flights

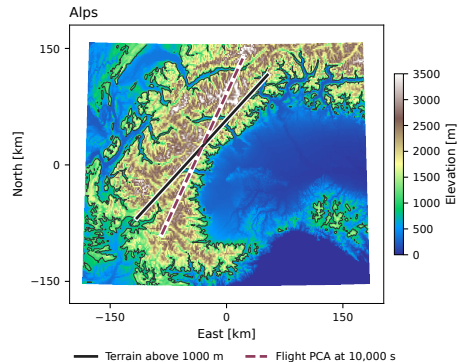
Discussion. Would local axes along the actual tracks change the regional comparison?

Every fourth cell of the 15-arc-second grid gives one-arc-minute spacing. We select elevations at or above 1000 metres without fitting the threshold to the flight PCA. WGS84 tangent coordinates use the box centre and zero projection height. Weights are cosine latitude to approximate cell area; elevation above the threshold adds no weight. Threshold sensitivity is descriptive. The box boundaries, curved chains and plateaux affect the direction. Flight boxes identify take-offs, while increments can leave them and use flight-centred tangent frames. This is an indicative regional comparison.

Alps: shared sector, with an approximately 18-degree offset

Discussion. How much of the offset reflects Alpine curvature, box truncation or route selection?

Both orientations occupy the northeast-southwest sector, but the flight covariance is more northward. A simple claim that flights run parallel to one Alpine axis misses the approximately eighteen-degree difference. The terrain angle spans 42.1 to 47.9 degrees over the three thresholds, so this sensitivity does not eliminate the offset. Wind and long-flight selection remain possible contributors; the figure neither isolates a terrain effect nor tests pilot skill.



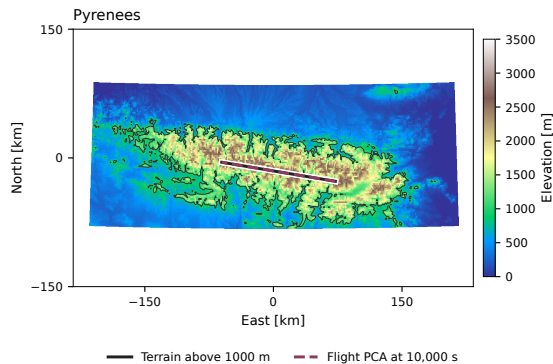
Terrain: 47.6° .

Flight PCA: 65.2° .

Separation: 17.6° .

46,336 flights support 10,000 s.

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.



Terrain: 170.5° .

Flight PCA: 170.5° .

Separation: less than 1° .

4,331 flights support 10,000 s.

ETOPO footprint above 1000 m; threshold sensitivity: $169.7\text{--}175.4^\circ$. Decimal agreement does not imply this physical precision.

Pyrenees: the long-lag spread follows the highland axis

Discussion. Does the alignment persist for fixed flights and for terrain sampled along their routes?

The near-parallel axes support a geographical association between displacement orientation and the range. They do not establish the causal contribution of terrain. Winds can align with the same range and pilots can select routes along it. The sensitivity span is not a sampling confidence interval. The comparison is made within the exact Pyrenean take-off box, but flight increments can extend beyond it.

Same population as the 10,000-s PCA

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

Mean: 1,006 m

Equal flight weights: 980 m.

Middle half of window means:
854 m–1,160 m.

Approximate altitude above sea level;
recorder datums are not harmonised.

Channel Coast: the reference height follows the PCA windows

Discussion. Is one representative height sufficient, or should wind be matched to the altitude and location of each flight window?

The mean is 1006.262 metres across 2013 exact PCA windows from 1680 flights, with the same equal-window weighting as the covariance. Each window altitude is a trapezoidal time average of cleaned GNSS z interpolated onto the existing 10-second grid. No gap is bridged. Equal weight per contributing flight gives 980.260 metres; the window-mean quartiles are 853.932 and 1159.549 metres. Short flights do not define this target because they cannot contribute at 10000 seconds. The raw GNSS vertical datum is not harmonised, so MSL is an approximation; these numbers do not imply metre-level physical accuracy. The quartile range describes heterogeneity, not uncertainty in the mean. Matching the target height does not yet match flight dates, instantaneous positions or local atmospheric conditions.

2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.

$$H = \Phi / g_0, \quad h = \frac{R_{\oplus} H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

Main comparison: all year, all hours vs. all year, 09–17 UTC.

April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

ECMWF/Copernicus ERA5 pressure levels, accessed through a pinned snapshot of Earthmover's public cloud-optimised ERA5 copy on AWS. Earthmover distributes this lossless copy; it is not a separate reanalysis. Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

ERA5 wind is interpolated to the measured flight altitude

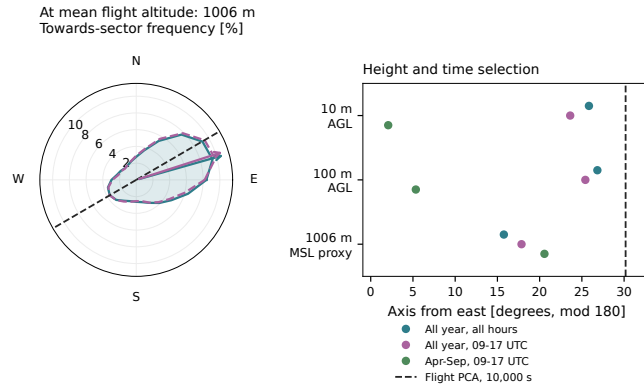
Discussion. How much of the surface-to-flight-height contrast reflects vertical shear, season and the flight conditions represented?

Use the same nine grid cells as the surface analysis: latitudes 48.75, 49.75, 50.75 degrees and longitudes minus 1.25, zero, 1.25 degrees. Geopotential height is Phi divided by 9.80665; geometric height uses spherical Earth radius 6371000 metres. A pressure level changes altitude from hour to hour. Linear interpolation acts on eastward and northward velocities, never direction angles. All 789048 cell-hour profiles bracket the main target above the model surface. Average components over time and with cosine-latitude cell weights, then reduce the towards-angle modulo 180 degrees for comparison with PCA. The annual all-hour selection has 87672 hours per cell; 09 through 17 UTC inclusive has 32877. The two warm-season selections have 43920 and 16470. UTC is fixed, without a daylight-saving correction. Source values, timestamps, coordinates, pressure levels, geopotential, surface geopotential and hashes are preserved losslessly for offline reuse.

At flight height, both annual references remain offset from PCA

Discussion. Why is the seasonal contrast larger near the surface?

At the mean flight altitude the all-year all-hour axis is 15.8 degrees and the 09–17 UTC axis is 17.9 degrees, against 30.2 degrees for flight PCA. The corresponding separations are 14.4 and 12.3 degrees. At 10/100 metres AGL the annual axes lie closer to PCA, but their warm-daytime references rotate further away. At flight altitude the warm-daytime axis is 20.6 degrees, giving a smaller 9.6-degree separation. The altitude-matched comparison therefore has a less close annual alignment and a smaller time-selection sensitivity than the surface comparison. These are descriptive contrasts, not formal tests of different angles. The rose uses weighted 15-degree sectors, excludes speeds below 0.5 m/s only from frequencies, and retains all hours in vector means. Teal is all-year all hours; plum is all-year daytime. Green appears only in the right-hand seasonal check.



Annual gaps at flight height: 14.4° for all hours, 12.3° for 09–17 UTC.

Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the 30.2° centred PCA axis.

At 1,006 m	Wind axis	PCA gap	ρ_w
All year, all hours	15.8°	14.4°	0.43
All year, 09–17 UTC	17.9°	12.3°	0.44
Apr–Sep, all hours	16.0°	14.2°	0.35
Apr–Sep, 09–17 UTC	20.6°	9.6°	0.37

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$ measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

Height and time checks preserve a broad orientation resemblance

Discussion. Does the remaining offset reflect flight selection and route choices, or change when weather is matched to actual flights?

The annual all-hour and daytime resultant fractions are 0.43 and 0.44; warm-season values are 0.35 and 0.37. These do not give confidence in a direction or the fraction of observations near it. At the lower quartile height, 868 profiles at one cell have no above-ground pressure level below the target. For the height sensitivity only, exclude these 868 hours from all nine cells at every tested height, including the mean-height reference: 86804 common hours remain per cell. Apply the same seasonal and daytime masks afterward. On common support the annual quartile-height axes range from 14.3 to 17.2 degrees for all hours and from 16.4 to 19.3 degrees for daytime. Equal-flight mean altitude changes the matched reference by less than 0.3 degrees, and common-hour selection changes the main annual reference by less than 0.1 degrees. These checks retain the offset but do not calibrate receiver datums or establish significance.

Region	Conclusion supported by this comparison
Pyrenees	Close terrain–flight alignment; a geographical association worth testing along tracks.
Alps	Same broad sector, with an approximately 18-degree offset from the mean highland axis.
Channel Coast	Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance.

A common constant wind shifts the mean displacement by $\tau\mathbf{w}$; centring removes it. Variable winds and route responses can still shape covariance.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

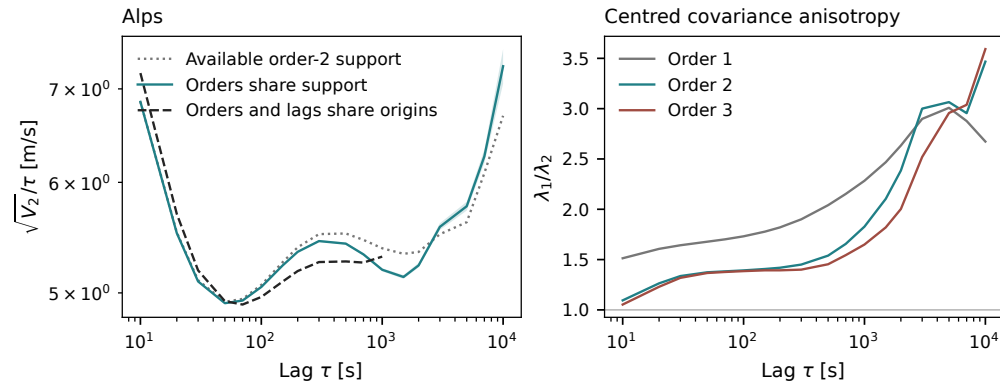
What the environmental comparisons establish

Discussion. Match flight times, heights and routes to wind and local terrain before attributing the ellipse or equipment contrasts to a cause.

The PCA concerns centred displacement spread, whereas the wind reference concerns a mean velocity. Their orientation resemblance alone cannot show that mean wind causes covariance anisotropy. Constant additive drift cancels when displacement is centred at fixed lag. Variable drift between flights, time-varying forcing and route decisions can alter covariance; their contributions need matched observations. The Pyrenean agreement is the strongest geographical alignment here. The coastal comparison at mean flight altitude retains a broadly similar sector across the four time selections, with offsets of about 10 to 14 degrees. Surface winds exhibit a larger seasonal change, so their annual near alignment should not be transferred to flight altitude. Equipment class is not measured pilot expertise, and this comparison cannot establish resistance to wind.

Thesis §3.5.2

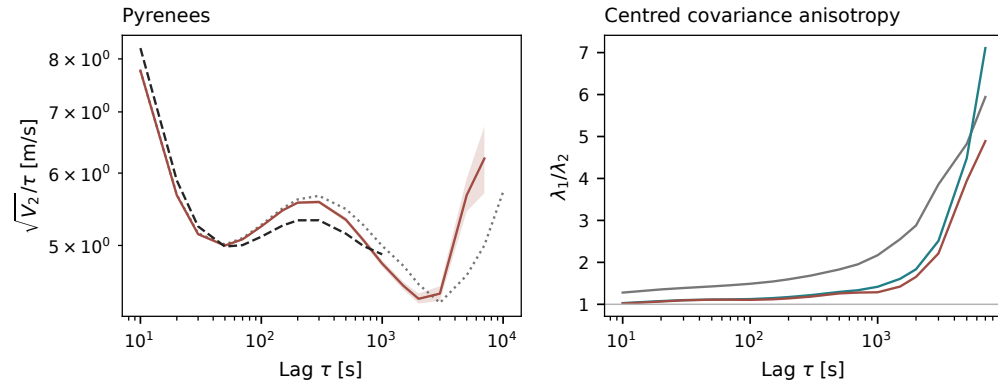
Regional changes of velocity and direction

Alps: velocity changes retain a directional structure

Equal flight weights. At 1000 s, matched orders retain 89,181 flights. Shading: pointwise 95% site-day intervals.

Discussion. Which part of the lag dependence persists with common origins and after changing difference order?

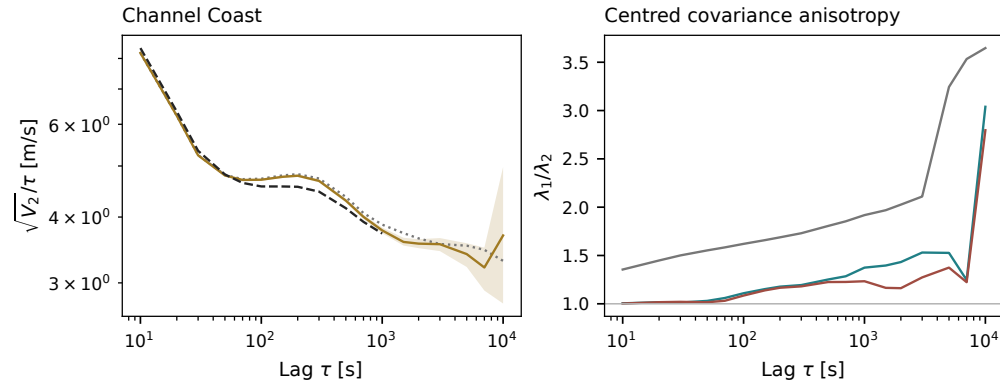
Left: available-order-two origins dotted, matched orders solid, and common origins through 1000 seconds dashed. Right: covariance anisotropy of orders one, two and three on identical origins. Unlike the preceding pooled-origin PCA, these curves give equal weight to flights. The full report also saves the pooled-origin sensitivity. Eighteen lags cover 10 to 10000 seconds. Matching all orders requires three times the lag; common origins use every ten-second origin supporting a 3000-second stencil. This changes the observations as well as flight membership.

Pyrenees: the long-lag change coincides with falling support

At 1000 s: 12,020 matched flights; at 10,000 s: only 3. Cells below eight flights are omitted from plots.

Discussion. How much of the late rise survives a comparison with fixed flights and origins?

The conventions are identical to the Alpine panel. At 1000 seconds the order-two eigenvalue ratio is 1.42, compared with 2.17 for ordinary increments. The sharp late changes need to be read together with the support figure. All numerical cells are retained in the report, including the final three-flight cell, but that cell is not plotted as a population result. The omitted 10000-second order-two ellipse is therefore intentional.

Channel Coast: smaller changes at 1000 s, with residual anisotropy

At 1000 s: 4,679 matched flights, $D_2 = 3.77$ m/s and covariance ratio 1.37.

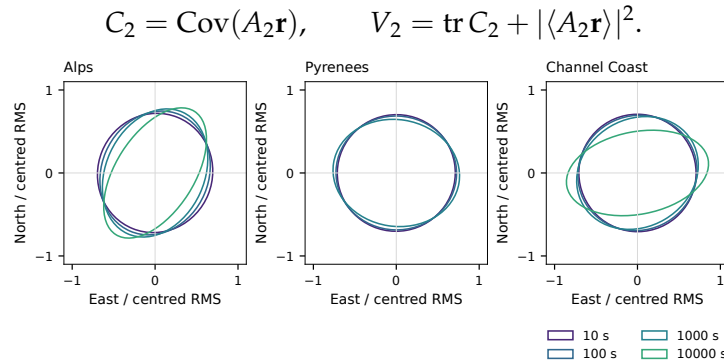
Discussion. Could route geometry and varying forcing explain the contrast that survives constant-velocity cancellation?

The coastal RMS change becomes lower than the mountain values over part of the lag range. It is higher than the Alpine value at ten seconds, so one amplitude ratio does not describe every scale. An added constant wind alone cannot explain a V_2 difference between otherwise identical populations. Variable wind, its effect on decisions, route geometry and selection remain possible. These boxes classify take-off locations, not exposure along the whole flight.

Second-difference ellipses retain orientation after drift cancellation

Discussion. Do the resolved axes align with independent terrain and wind information?

The four colours match the exact PCA display lags. Ellipses describe centred spread and omit both amplitude and mean. Near-circular ellipses, especially the coastal ten-second one, do not resolve a useful direction. At 1000 seconds the order-two axes are approximately 65 degrees in the Alps, 171 degrees in the Pyrenees and 31 degrees on the coast, counterclockwise from east. Axes are modulo 180 degrees and do not supply signed wind directions. The preceding independent comparisons provide regional references; they do not measure terrain or weather exposure along individual tracks, and the coastal mean-height wind reference retains a 10–14-degree offset across the tested time selections.

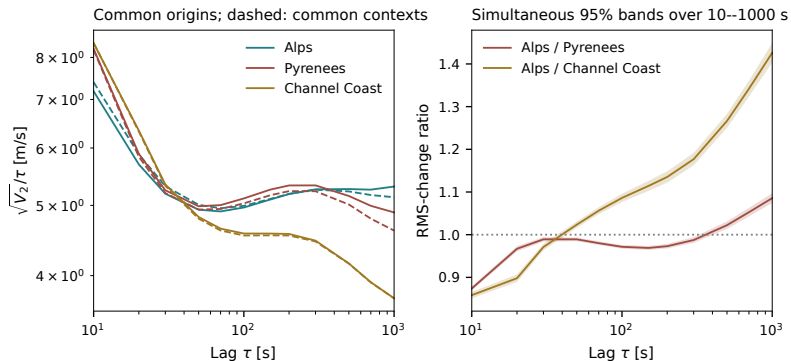


Matched-order origins, equal flight weights; each ellipse is normalised by its own centred RMS.

The regional contrast persists in a common-context control

Discussion. Which part survives matching actual weather, sites and pilot identity?

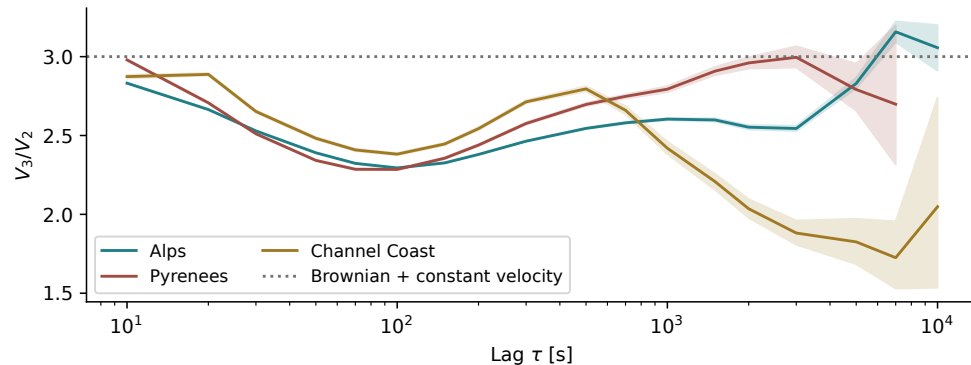
The context control standardises task, retained-duration band, season period and equipment, with at least ten flights per region per retained stratum. Target stratum mass is proportional to the smallest regional count. It retains 65537 Alpine, 9933 Pyrenean and 4366 coastal flights. The 1000-second RMS changes are 5.13, 4.62 and 3.72 m/s. These dashed point estimates have no calibrated intervals. The shaded ratio intervals instead apply to the unstandardised common-origin population and use 500 site-day bootstrap replicates jointly over the supported lag grid. The Alpine-to-coastal ratio there is 1.426 with a simultaneous band from 1.403 to 1.449 at 1000 seconds. No equivalence margin was specified, so a non-significant difference would not establish equality.



Common origins through 1000 s. Dashed: 23 shared context strata, descriptive. Shading: simultaneous 95% bands for the unstandardised ratios.

Thesis §3.5.2

Third differences and the distribution of changes



Matched origins and flight weights; pointwise 95% site-day intervals. Dotted: Brownian motion plus constant velocity.

The ratio between orders changes with the lag

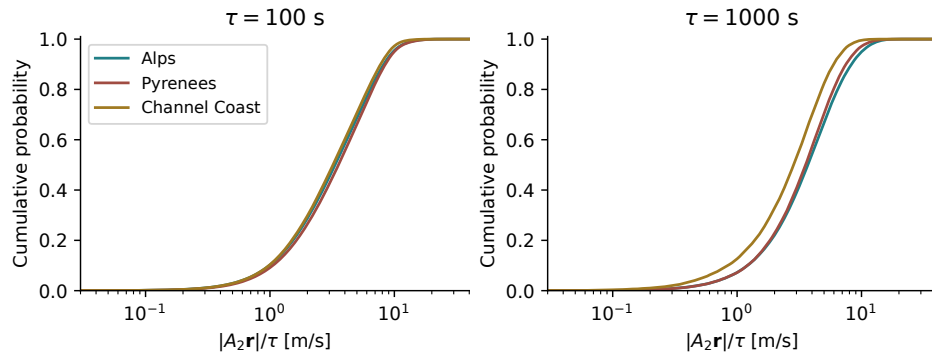
Discussion. Which scale interval could satisfy both the slope and amplitude predictions of one model?

V_3/V_2 is approximately 2.29 in both mountain regions at 100 seconds; at 1000 seconds it is 2.60 in the Alps, 2.79 in the Pyrenees and 2.42 on the coast. The lag dependence is itself diagnostic. V_3 is not an independent sample and does not automatically isolate still-air dynamics. Its longer support and larger sensitivity to position noise must accompany any interpretation. No Hurst estimate is validated by these curves.

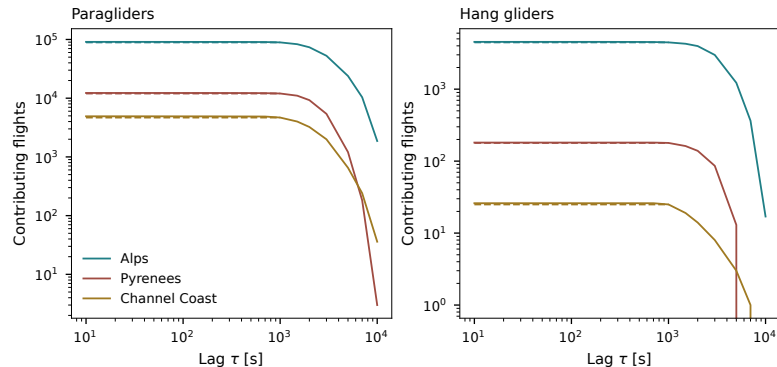
The distribution distinguishes frequent small changes from large events

Discussion. Where on the trajectories do the largest contributions occur, and which verified flight phases contain them?

At 1000 seconds the upper tenth of changes contributes about 43–47 percent of the squared-change signal in the Alps, 40–44 percent in the Pyrenees and 38–43 percent on the coast. Those ranges bound the unresolved histogram boundary bin; they are not sampling confidence intervals. The result measures concentration of this observable. It does not label turbulence or a particular flight phase. Phase conditioning and independent wind attribution have not been performed in this extension.



Equal-flight CDFs on matched-order support. Full report: all four display lags, zero and overflow mass, and quantile bin bounds.

Order-three support limits the long-lag regional comparison

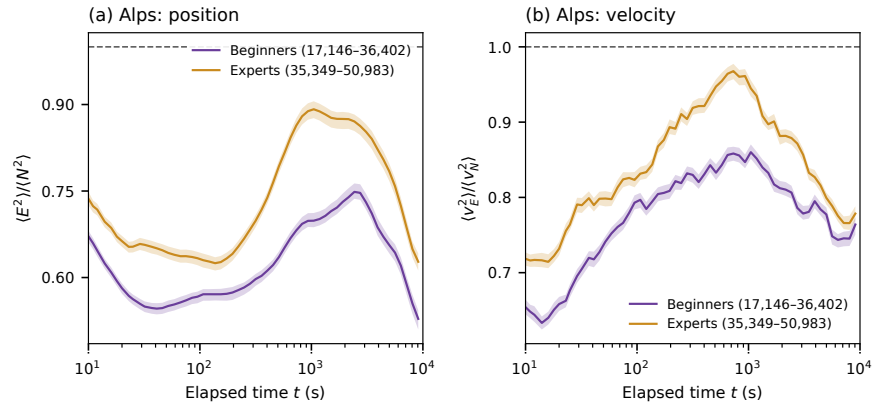
Solid: orders share 3τ support. Dashed: origins common through 1000 s. All eligible regional flights contribute where supported.

Discussion. Which conclusions remain supported in both disciplines and across the same range of scales?

The full regional measurement includes 108189 paraglider flights and 4747 hang-glider flights. At 10000 seconds, matched paraglider counts are 1865, 3 and 36 for Alps, Pyrenees and Channel Coast; hang-glider counts are 17, zero and zero. Coastal hang gliders number only 26 even at short lags. The report retains their estimates and suppresses intervals below twenty contributing site-day groups. No common three-region equipment/context strata survive for hang gliders, so that standardisation is not claimed for this discipline.

Thesis §3.5.3

Terrain, equipment and wind

**Alps: C/D/CCC has smaller coordinate-moment imbalance**

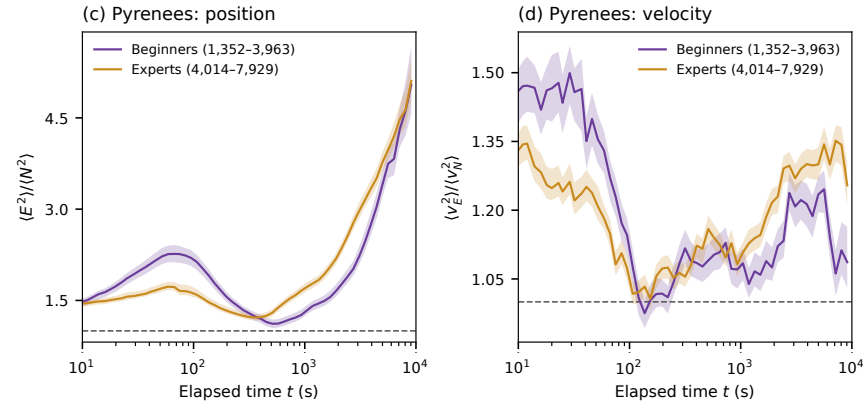
Discussion. Does the ordering persist after matching site, day, task and duration?

The fresh Alpine position and velocity ratios stay closer to unity for C/D/CCC over the supported grid. This is a descriptive ensemble observation. It does not establish individual isotropy, greater skill or a causal equipment effect. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

Pyrenees: the ordering of the two groups reverses

Discussion. What changes between the early and later portions of the curves?

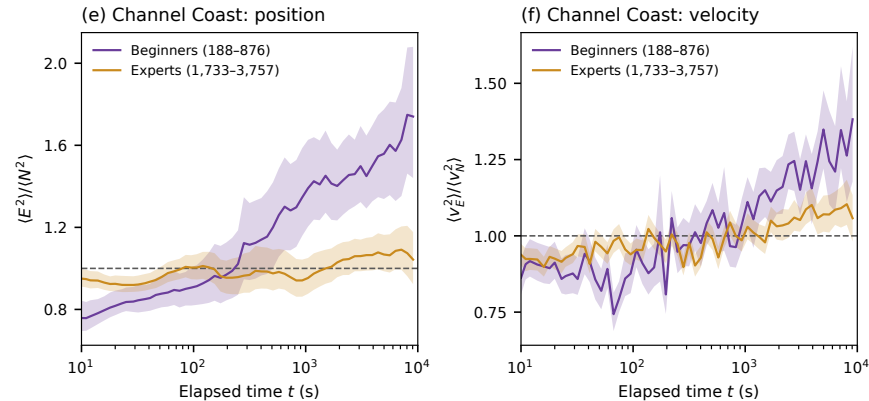
The Pyrenean curves do not support a uniform expert-proxy advantage. At late times both position ratios exceed five, while the velocity ordering also changes. Adjacent plotted times are correlated observations, not independent tests. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



Channel Coast: the late position contrast is pronounced

Discussion. Can matching conditions separate directional route choice from wind response?

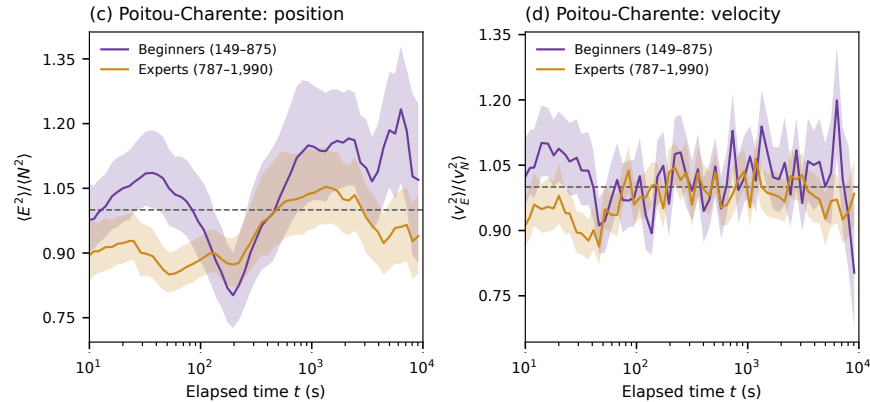
Near 9086 seconds the position ratios are about 1.74 for A/B and 1.04 for C/D/CCC, supported by 188 and 1733 paired flights. These are raw ensemble second moments and the samples differ strongly. Coastal cliffs and valleys prevent a wind-only interpretation. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



Poitou-Charente: the group curves cross

Discussion. Does this low-relief launch box reproduce the coastal observation?

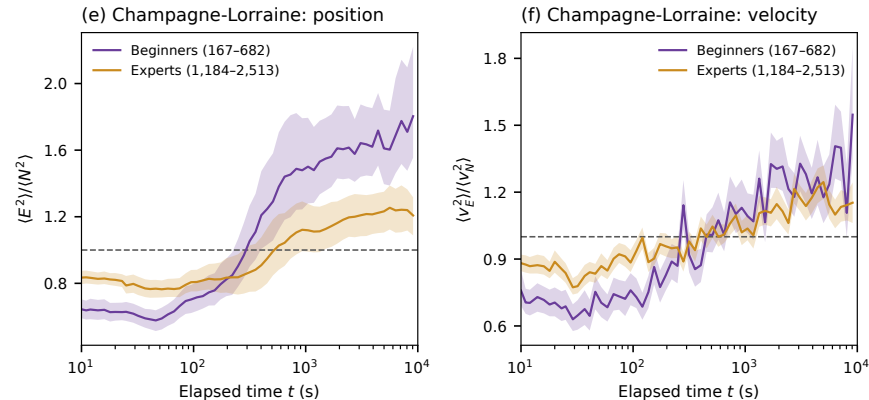
The crossing curves and broad overlapping bands do not support a general group ordering. These broad launch-location boxes are neither DEM-verified plains nor controls matched on weather. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity

Discussion. Which parts of the coastal hypothesis transfer to this separate region?

A later position contrast appears, but the C/D/CCC ratio also rises above one. Region, route mixture, weather and declining support must be separated before claiming the same physical cause. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



Hypothesis to discuss

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

Evidence limiting that interpretation

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

The environmental interpretation remains a hypothesis

Discussion. What observation would distinguish this hypothesis from different sites, tasks, weather or route mixtures?

Preserve the proposed explanation as a useful hypothesis rather than silently removing it or presenting it as established. Similar curve shape does not by itself establish a common cause or an unavoidable terrain constraint. Wind and terrain can interact, so the coast/mountains split does not identify their separate effects.

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Upwind progress, air-relative performance, choice of route and geographic moment balance are different outcomes.

A first comparison would match overlapping site–day, duration and task support; then separate mean drift, centred variability and individual route orientation. Regional references show terrain alignment and a remaining coastal wind–PCA offset at flight altitude; local track-matched observations are still needed.

Define the target before calling it resistance to wind

Discussion. Which target and matched comparison should be the first priority?

The site–day bootstrap adjusts one uncertainty structure; it does not perform matching or remove confounding. Where data permit, the same pilot on different equipment could help assess selection, but weather and task still require control. Differentiating velocity removes a constant wind term, but not its variation along the trajectory or the pilot’s manoeuvres.